

Group Assignment

This assignment must be completed in groups made up of 4 to 5 students following the usual criteria in Nova:

- Nationality: Including at least two nationalities;
- Gender: Including at least one member from each gender;
- Background: Including at least one member with different academic backgrounds (e.g. different degrees).

The assignment must be handed in through Moodle in pdf. Font size cannot be less than 11pt, margins cannot be smaller than 1", and line spacing cannot be smaller than 1.5. The first page must be a cover page with your name and student ID numbers.

The assignment requires downloading data from WRDS. You can create a WRDS account through this link: [WRDS-access](#) using the class code 628d431c.

We are interested in understanding the relationship between the minimum tick and volume. The assignment relies on an experiment conducted by the SEC in 2016 in which the minimum tick of a random set of firms was changed. You can read about the experiment in [Albuquerque et al. \(2020\)](#) and [here](#). You can find the data about which stocks were treated and which were control in the file *Treatmentcontrollist* available through Moodle.

The assignment must have four sections:

Hypotheses development (max. 2 pages, 2 points): you need to create two hypotheses that link minimum tick and volume. You can consider a different relation for different types of firms, time periods, etc.

Empirical methodology (max. 2 pages, 4 points): you need to present your empirical methodology (e.g., difference-in-differences, IV, etc.), justify why you chose that particular methodology, and explain which controls you would use and why.

Data (max. 4 pages including a table, 6 points): You need to describe in detail how you constructed the final sample. This section must include how you construct variables as well as how you selected observations (e.g. I disregard stocks with missing volume data). It must include a table with summary statistics of all the variables that you will use (and only those). You can use Table 1 in Albuquerque et al. (2020) as a guide.

Results (max. 4 pages including one table, 8 points): You must report your results in a Table with the corresponding explanatory note (use any table in Albuquerque et al. (2020) as a guide). Then, you need to interpret those coefficients in terms of your hypothesis.

In addition to those four sections, the thesis must have a page of references and a title page, which must include the names and student numbers of all group members.

You must submit the script to obtain the results in addition to the PDF with the answers. You can use any software you want, but we will only provide support for Stata. The script will not be graded.

Notes:

- Use data from January 1, 2016 to April 30, 2019
- Results might differ from Albuquerque et al. 2020
- Do not condition on small and quoted spread when computing the summary statistics but condition on treatment as they do (control, group 1, etc.)
- Include in the summary statistics only the variables you will use and not all variables in Albuquerque et al. 2020

If you have any questions, you can email Fabian Waßmann at fabian.wassmann@novasbe.pt or schedule office hours with him through Moodle.